Horizontal launches

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find flight time from vertical motion and then horizontal range.

SUCCESS CRITERIA - I can...

1. Separate vertical and horizontal givens.
2. Find or use fall time before calculating range.
3. Check units and interpret the landing point.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “horizontal” and “vertical” to explain your answer. Sentence starters are optional.

horizontal: along a level line, usually the x direction

vertical: up and down, usually the y direction

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

At what altitude h/R is a spherical planet’s field **1/9** of its surface value? Repeat for **1/25**.

Word help: altitude = height above a reference surface

Sentence starter: My design includes ... . It does what is needed by ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $h = \frac{1}{2}gt^2$; range = $v_x t$.
Horizontal launch; h is the positive drop, t is fall time; g = **10 m/s²**. CPA may use the supplied time.

Worked reasoning: A horizontal launch falls **20 m** with g=**10 m/s²**. From **20**=**5** t², t=**2 s**. At horizontal speed **3 m/s**, range=**6 m**.

Your turn: Draw a diagram labeling height and range. Number the two solution steps to **show why time comes first**.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

On the diagram, label height h, horizontal range and initial horizontal velocity. **Which direction** determines the fall time?

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

04 Apply the physics

In a launch simulation, a ball leaves a **45 m** cliff horizontally at **15 m/s**.
1. **Label height** and range on the diagram.
2. **Find the flight time first**, then **the range**.
3. Use g = **10 m/s²** and ignore air resistance.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Two platforms have heights **5** and **20 m**. Supplied fall times are **1** and **2 s**. At the same launch speed, does quadrupling height quadruple range? Explain.

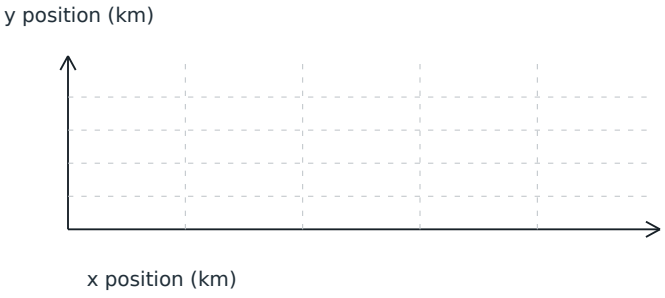
Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A simplified calculation predicts position (**-60**, **80**) km with bounds ±**6 km** in x and ±**8 km** in y. **Write a short briefing giving** both intervals and two reasons it is not a real asteroid collision forecast.
Label the given and predicted quantities on the diagram.

Sentence starter: I predict … . I used … to make this prediction because … .



Marzano check - circle: 1 2 3

07 Exit check

From a **20 m** cliff, horizontal launch speed is **8 m/s**. **Find t**, range and final vertical velocity. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Sentence starter: I found … [unit]. This value means … in this situation.

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A horizontal launch at **12 m/s** must land **18 m** from the base. **Find launch height** and impact speed using g=**10 m/s²**. **Label the diagram** with the known quantities before calculating. Ignore air resistance; use g = **10 m/s²** when needed.

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3